Alpha Beta Synthetic Sampling Methods Study

HELIX FORGE LAB / p04-full-007230 / RUN R184 / 2026-07-13

ALPHA F1	PRECISION	RECALL	BETA F1
0.873	0.881	0.865	0.846

Decision context

This report records a bounded synthetic sampling evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

Lineage

Upstream evidence: p04-full-000015 (models/model-cards/model-reviews/alpha-beta/calibration/tables/alpha-beta-calibration-metrics-v01.csv).

Slice	Records	Precision	Recall	F1
slice-01	176	0.883	0.881	0.882
slice-02	205	0.865	0.857	0.861
slice-03	190	0.882	0.870	0.876
slice-04	201	0.864	0.883	0.873
slice-05	225	0.881	0.859	0.870